WORKED SOLUTION

STEP II 2009, Question 4

Pure • Polynomial Divisibility

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/09-s2-q4

The key idea

Divisibility by $(x-1)^5$ is best handled by writing it out: $\mathrm{p}(x) = 1 + (x-1)^5\mathrm{q}(x)$ for some polynomial q. Differentiating shows that a repeated factor of $\mathrm{p}(x) - 1$ survives, one power lower, in $\mathrm{p}'(x)$. In the final part this pins down p' almost completely, so we find p by integrating p' rather than by wrestling with ten unknown coefficients.

Solution

Since $\mathrm{p}(x) - 1$ is divisible by $(x-1)^5$ and has degree 9, we can write

$$\mathrm{p}(x) = 1 + (x-1)^5\,\mathrm{q}(x),$$

where q is a polynomial of degree 4.

Part (i)

Putting $x = 1$ kills the second term, so

$$\mathrm{p}(1) = 1.$$

Part (ii)

Differentiating with the product rule,

$$\begin{aligned}\mathrm{p}'(x) &= 5(x-1)^4\,\mathrm{q}(x) + (x-1)^5\,\mathrm{q}'(x)\\ &= (x-1)^4(5\mathrm{q}(x) + (x-1)\mathrm{q}'(x)).\end{aligned}$$

The bracket is a polynomial, so $\mathrm{p}'(x)$ is exactly divisible by $(x-1)^4$, as required.

Part (iii)

Exactly the same argument with $x+1$ in place of $x-1$ applies: writing $\mathrm{p}(x) = -1 + (x+1)^5\,\mathrm{r}(x)$ gives $\mathrm{p}(-1) = -1$ and shows that $\mathrm{p}'(x)$ is divisible by $(x+1)^4$.

So $\mathrm{p}'(x)$ is divisible by both $(x-1)^4$ and $(x+1)^4$.

Ask yourself Why does that make $\mathrm{p}'(x)$ divisible by the product $(x-1)^4(x+1)^4$?

Write $\mathrm{p}'(x) = (x-1)^4\mathrm{s}(x)$. Every root of $(x+1)^4$ is -1, which is not a root of $(x-1)^4$, so the factor $(x+1)^4$ must divide $\mathrm{s}(x)$. (Equivalently, the linear factor $x+1$ appears four times in p', and none of those copies can be used up by $(x-1)^4$.) Hence $(x-1)^4(x+1)^4 = (x^2-1)^4$ divides $\mathrm{p}'(x)$.

Now p' has degree 8, the same as $(x^2-1)^4$, so the quotient is a non-zero constant:

$$\mathrm{p}'(x) = k(x^2-1)^4 = k\left(x^8 - 4x^6 + 6x^4 - 4x^2 + 1\right).$$

Integrating,

$$\mathrm{p}(x) = k\,\mathrm{F}(x) + c, \qquad \mathrm{F}(x) = \frac{x^9}{9} - \frac{4x^7}{7} + \frac{6x^5}{5} - \frac{4x^3}{3} + x.$$

Notice that F is odd, so $\mathrm{F}(-1) = -\mathrm{F}(1)$. The two values we know give

$$k\mathrm{F}(1) + c = \mathrm{p}(1) = 1, \qquad -k\mathrm{F}(1) + c = \mathrm{p}(-1) = -1.$$

Adding, $c = 0$; subtracting, $k\mathrm{F}(1) = 1$. With a common denominator of 315,

$$\mathrm{F}(1) = \frac{35 - 180 + 378 - 420 + 315}{315} = \frac{128}{315}, \qquad \text{so} \qquad k = \frac{315}{128}.$$

Therefore

$$\mathrm{p}(x) = \frac{1}{128}\left(35x^9 - 180x^7 + 378x^5 - 420x^3 + 315x\right).$$

We should also check that this polynomial really works, since so far we have only shown that any valid p must have this form. It has degree 9. Also $\mathrm{p}(1) - 1 = 0$ and $\mathrm{p}'(x) = \frac{315}{128}(x-1)^4(x+1)^4$, so the first four derivatives of $\mathrm{p}(x) - 1$ all vanish at $x = 1$; thus $x = 1$ is a root of $\mathrm{p}(x) - 1$ of multiplicity at least 5, i.e. $(x-1)^5$ divides $\mathrm{p}(x) - 1$. Since p is odd, $\mathrm{p}(x) + 1 = -(\mathrm{p}(-x) - 1)$, which is divisible by $(-x-1)^5 = -(x+1)^5$. So both conditions hold, and this p is the unique answer.

Checks

- The coefficients $35, -180, 378, -420, 315$ sum to 128, confirming $\mathrm{p}(1) = 1$.
- Dividing $\mathrm{p}(x) - 1$ repeatedly by $x - 1$ with exact fractions gives remainder zero exactly five times, and likewise $\mathrm{p}(x) + 1$ by $x + 1$.
- p is odd, as it must be: if p works then so does $-\mathrm{p}(-x)$, and the answer is unique.

Common slips

- Trying to find all ten coefficients of p directly from ten simultaneous equations. It works in principle, but integrating p' is far quicker.
- Forgetting the constant of integration, or not using both $\mathrm{p}(1) = 1$ and $\mathrm{p}(-1) = -1$ to fix it and k.

Beyond STEP

The fact behind part (ii) is that a is a root of multiplicity m of f exactly when $\mathrm{f}(a) = \mathrm{f}'(a) = \cdots = \mathrm{f}^{(m-1)}(a) = 0$. Prescribing values and derivatives at points like this is *Hermite interpolation*, and the polynomial found here is a "smoothstep" curve, rising from -1 to 1 as flatly as possible at both ends, which is used in computer graphics. The argument that coprime factors divide a product is a piece of ring theory: polynomials over a field have unique factorisation, just like the integers. For more, see Allenby, *Rings, Fields and Groups*.

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